

Example 2

Below are the scores of two cricketers in 10 innings. Find who is the more “consistent scorer” by evaluating their relative dispersion.

Cricketer A	204	68	150	30	70	95	60	76	24	19
Cricketer B	99	190	130	94	80	89	69	85	65	40

Solution

Using standard sample statistical indicators for the datasets ($n = 10$ innings):

1. Analysis for Cricketer A:

$$\sum x_A = 796 \Rightarrow \bar{x}_A = \frac{796}{10} = 79.60$$

$$\sum x_A^2 = 93758 \Rightarrow s_A = \sqrt{\frac{93758 - \frac{(796)^2}{10}}{9}} = \sqrt{3391.6} \approx 58.24$$

$$\text{C.V}_A = \left(\frac{58.24}{79.60} \right) \times 100\% \approx 73.17\%$$

2. Analysis for Cricketer B:

$$\sum x_B = 941 \Rightarrow \bar{x}_B = \frac{941}{10} = 94.10$$

$$\sum x_B^2 = 103769 \Rightarrow s_B = \sqrt{\frac{103769 - \frac{(941)^2}{10}}{9}} = \sqrt{1691.21} \approx 41.12$$

$$\text{C.V}_B = \left(\frac{41.12}{94.10} \right) \times 100\% \approx 43.70\%$$

Conclusion: The Coefficient of Variation of Cricketer A (73.17%) is significantly greater than the Coefficient of Variation of Cricketer B (43.70%). A lower coefficient of variation denotes greater stability; hence, we conclude that **Cricketer B is a more consistent scorer.**

Measures of Shape

Skewness

Before discussing the concept of skewness, an understanding of the concept of symmetry is essential.

A plot of frequency against class mark joined with a smooth curve can help us to visually assess the symmetry of a distribution.

Usually symmetry is about the central value. Symmetry is said to exist if the smoothed frequency polygon of the distribution can be divided into two identical halves which are mirror images of each other. Skewness on the other hand means lack of symmetry and it can be positive or negative.

Basically, if the distribution has a tail on the right, (See figure below), then the distribution is positively skewed Eg Most students having very low marks in an examination.

Generally, for any set of values, the **moment coefficient of skewness** (α_3 or β_1) is given by:

$$\alpha_3 = \frac{\mu_3}{\sigma^3} \quad \text{or} \quad \alpha_3 = \frac{\sum (x_i - \bar{x})^3}{n \cdot s^3}$$

where s (or σ) is the standard deviation, and μ_3 is the third central moment. It is worth noting that:

- If $\alpha_3 < 0$, the distribution is **negatively skewed** (skewed left).
- If $\alpha_3 > 0$, the distribution is **positively skewed** (skewed right).
- If $\alpha_3 = 0$, the distribution is **perfectly symmetrical** (normal).

Other standard metrics include **Karl Pearson's coefficient of skewness** (S_k), **Bowley's coefficient of skewness** (B_k), and **Kelley's coefficient of skewness** (K_k).

1. Karl Pearson's Coefficient of Skewness

The Karl Pearson's coefficient of skewness is based upon the divergence of the mean from the mode in a skewed distribution. Recall the empirical relation between mean, median, and mode, which states that for a moderately asymmetrical distribution, we have:

$$\text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median})$$

Hence, Karl Pearson's coefficient of skewness is defined by:

$$S_k = \frac{\text{Mean} - \text{Mode}}{s} \quad \text{or, alternatively,} \quad S_k = \frac{3(\text{Mean} - \text{Median})}{s}$$

2. Bowley's Coefficient of Skewness

The Bowley's coefficient of skewness is based entirely on quartiles. For a perfectly symmetrical distribution, the lower quartile (Q_1) and upper quartile (Q_3) are equidistant from the median (Q_2):

$$B_k = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

where Q_k is the k -th quartile.

3. Kelley's Coefficient of Skewness

Kelley's coefficient of skewness is based on the 10th (P_{10}) and 90th (P_{90}) percentiles, ignoring only 10% of observations on each extreme. This provides an improvement over Bowley's coefficient, which leaves out 25% of observations on each tail:

$$K_k = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

where P_k represents the k -th percentile (note that the median $Q_2 = P_{50}$).

Interpreting Skewness (Bulmer's Rule of Thumb)

If the coefficient of skewness is positive, the data are skewed right (longer right tail). If negative, the data are skewed left (longer left tail). Bulmer, M. G., *Principles of Statistics* (Dover, 1979) suggests this practical rule of thumb:

- If skewness is **less than** -1 or **greater than** $+1$, the distribution is **highly skewed**.
- If skewness is **between** -1 and -0.5 or **between** $+0.5$ and $+1$, the distribution is **moderately skewed**.
- If skewness is **between** -0.5 and $+0.5$, the distribution is **approximately symmetric**.

Worked Example & Questions

Given the distribution relating to the capital size of 285 companies ($N = 285$):

Capital (Ksh lacs)	1-5	6-10	11-15	16-20	21-25	26-30	31-35
No. of Companies (f)	20	27	29	38	48	53	70
Cumulative Freq. (CF)	20	47	76	114	162	215	285

Part 1: Verification of Bowley's Coefficient of Skewness

- **Quartile 1 (Q_1):** Position = $\frac{285}{4} = 71.25$. Falls in class 10.5 – 15.5.

$$Q_1 = 10.5 + \left(\frac{71.25 - 47}{29} \right) \times 5 \approx 14.68$$

- **Median (Q_2):** Position = $\frac{285}{2} = 142.5$. Falls in class 20.5 – 25.5.

$$Q_2 = 20.5 + \left(\frac{142.5 - 114}{48} \right) \times 5 \approx 23.47$$

- **Quartile 3 (Q_3):** Position = $\frac{3(285)}{4} = 213.75$. Falls in class 25.5 – 30.5.

$$Q_3 = 25.5 + \left(\frac{213.75 - 162}{53} \right) \times 5 \approx 30.38$$

- **Bowley's Coefficient (B_k):**

$$B_k = \frac{30.38 + 14.68 - 2(23.47)}{30.38 - 14.68} = \frac{45.06 - 46.94}{15.70} = \frac{-1.88}{15.70} \approx -0.12$$

Interpretation: Since -0.12 lies between -0.5 and $+0.5$, the distribution is **approximately symmetric**.

Part 2: Solution for Karl Pearson's and Kelley's Coefficients

To calculate these metrics, we must first establish the midpoints (x), mean ($\bar{x}$), standard deviation (s), mode, and relevant extreme percentiles (P_{10}, P_{90}).

1. Summary Parameters

- **Mean ($\bar{x}$):** $\sum fx = (20 \times 3) + (27 \times 8) + (29 \times 13) + (38 \times 18) + (48 \times 23) + (53 \times 28) + (70 \times 33) = 6235$.

$$\bar{x} = \frac{6235}{285} \approx 21.88$$

- **Standard Deviation (s):** $\sum fx^2 = 162235$.

$$s = \sqrt{\frac{162235}{285} - (21.877)^2} = \sqrt{569.25 - 478.61} = \sqrt{90.64} \approx 9.52$$

- **Mode:** The modal class is 30.5 – 35.5 ($f = 70$). $\Delta_1 = 70 - 53 = 17$, $\Delta_2 = 70 - 0 = 70$.

$$\text{Mode} = 30.5 + \left(\frac{17}{17 + 70} \right) \times 5 = 30.5 + 0.98 = 31.48$$

2. Karl Pearson's Coefficient of Skewness (S_k)

Using the mode formula method:

$$S_k = \frac{\bar{x} - \text{Mode}}{s} = \frac{21.88 - 31.48}{9.52} = \frac{-9.60}{9.52} \approx -1.01$$

Alternatively, using the median formula configuration:

$$S_k = \frac{3(\bar{x} - \text{Median})}{s} = \frac{3(21.88 - 23.47)}{9.52} = \frac{-4.77}{9.52} \approx -0.50$$

Interpretation: Karl Pearson's index reveals a negative skewness. Depending on the method chosen, it spans between **moderately skewed** (-0.50) to **highly skewed** (-1.01) to the left.

3. Kelley's Coefficient of Skewness (K_k)

- **Percentile 10 (P_{10}):** Position = $0.10 \times 285 = 28.5$. Class: 5.5 – 10.5.

$$P_{10} = 5.5 + \left(\frac{28.5 - 20}{27} \right) \times 5 \approx 7.07$$

- **Percentile 90 (P_{90}):** Position = $0.90 \times 285 = 256.5$. Class: 30.5 – 35.5.

$$P_{90} = 30.5 + \left(\frac{256.5 - 215}{70} \right) \times 5 \approx 33.46$$

- **Kelley's Skewness (K_k):**

$$K_k = \frac{33.46 + 7.07 - 2(23.47)}{33.46 - 7.07} = \frac{40.53 - 46.94}{26.39} = \frac{-6.41}{26.39} \approx -0.24$$

Interpretation: Kelley's coefficient yields a value of -0.24 . Because this falls within the boundary range of -0.5 to $+0.5$, the distribution can be interpreted as **approximately symmetric**.

Kurtosis

It measures the peakedness of a distribution. If the values of x are very close to the mean, the peak is very high and the distribution is said to be Leptokurtic. On the other hand if the values of x are very far away from the mean, the peak is very low and the distribution is said to be Pletykurtic. Finally if x values are at a moderate distance from the mean then the peak is moderate and the distribution is said to be mesokurtic..

Generally, for a set of values, the **moment coefficient of kurtosis** (α_4 or β_2) is a measure of the peakedness or flatness of a distribution relative to a normal distribution. It is defined mathematically by:

$$\alpha_4 = \frac{\mu_4}{\sigma^4} \quad \text{or} \quad \alpha_4 = \frac{\sum (x_i - \bar{x})^4}{n \cdot s^4}$$

where $\bar{x}$ is the arithmetic mean, s is the standard deviation, and μ_4 is the fourth central moment.

Interpretation Boundaries:

- If $\alpha_4 > 3$, the distribution is **leptokurtic** (sharply peaked with heavy tails).
- If $\alpha_4 < 3$, the distribution is **platykurtic** (flat-topped with thin tails).
- If $\alpha_4 = 3$, the distribution is **mesokurtic** (perfectly normal peaking).

Example

Find the moment coefficient of skewness (α_3) and kurtosis (α_4) for the following dataset: \ 5, 6, 7, 6, 9, 4, 5

Solution Matrix

Given sample size $n = 7$. \ The arithmetic mean is computed as:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{5 + 6 + 7 + 6 + 9 + 4 + 5}{7} = \frac{42}{7} = 6.0$$

x_i	Deviation $(x_i - \bar{x})$	Squared $(x_i - \bar{x})^2$	Cubic $(x_i - \bar{x})^3$	Quartic $(x_i - \bar{x})^4$
5	-1	1	-1	1
6	0	0	0	0
7	1	1	1	1
6	0	0	0	0
9	3	9	27	81
4	-2	4	-8	16
5	-1	1	-1	1
Total	0	16	18	100

The standard deviation (s) is given by:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = \sqrt{\frac{16}{7}} \approx 1.5119$$

1. Coefficient of Skewness (α_3)

$$\alpha_3 = \frac{\sum (x_i - \bar{x})^3}{n \cdot s^3} = \frac{18}{7 \times (1.5119)^3} = \frac{18}{7 \times 3.4563} = \frac{18}{24.1941} \approx 0.744$$

Interpretation: Since $\alpha_3 = 0.744$ is positive, the distribution is **moderately skewed to the right**.

2. Coefficient of Kurtosis (α_4)

$$\alpha_4 = \frac{\sum (x_i - \bar{x})^4}{n \cdot s^4} = \frac{100}{7 \times (1.5119)^4} = \frac{100}{7 \times 5.2245} = \frac{100}{36.5715} \approx 2.734$$

Interpretation: Since $\alpha_4 = 2.734$, which is less than 3, the distribution is **platykurtic** (slightly flatter than a standard normal curve).